

Vehicle Dynamics with Dugoff Tire Model

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Abstract

This document presents the transition from a kinematic bicycle model to a dynamic vehicle model incorporating the Dugoff tire model with wheel rotational dynamics. We derive the complete force-based dynamics, normal load distribution with load transfer, slip ratio computation, friction-limited tire forces, and wheel angular velocity integration. The hybrid torque-slip model enables realistic simulation of traction limits, wheel spin, wheel lock, and combined longitudinal-lateral dynamics for autonomous mining haul truck applications.

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1 Introduction

1.1 Motivation for Dynamic Model

The kinematic bicycle model assumes:

- **No slip:** Pure rolling condition at all times
- **Unlimited traction:** Tire forces can be arbitrarily large
- **Instantaneous response:** No force buildup dynamics

These assumptions break down when:

- Vehicle operates on low-friction surfaces (gravel, wet haul roads)
- Emergency braking or aggressive acceleration
- Combined maneuvers (braking while turning)
- Realistic traction control and ABS simulation required

The Dugoff tire model addresses these by:

- Computing friction-limited forces based on normal load
- Modeling slip ratios (longitudinal and lateral)
- Capturing smooth saturation from linear to sliding regimes

1.2 Model Architecture Evolution

We present three levels of modeling fidelity:

Kinematic Model (V1):

$$\text{Actuator} \rightarrow \text{Drive} \rightarrow \text{Velocity} \rightarrow \text{Position/Yaw} \quad (1)$$

Torque-Only Dynamic Model (V2):

$$\text{Actuator} \rightarrow \text{Drive} \rightarrow \boxed{\text{Tire Forces (friction-limited)}} \rightarrow \text{Acceleration} \rightarrow \text{Velocity} \quad (2)$$

Hybrid Torque-Slip Model (V2.5):

$$\text{Actuator} \rightarrow \boxed{\text{Wheel Dynamics}} \rightarrow \boxed{\text{Slip}} \rightarrow \boxed{\text{Dugoff Forces}} \rightarrow \text{Vehicle Dynamics} \quad (3)$$

The hybrid model adds wheel rotational dynamics, enabling natural slip development, wheel spin, and wheel lock simulation.

1.3 Model Comparison Summary

Feature	Kinematic	Torque-Only	Hybrid
Traction limit	Unlimited	μF_z	μF_z
Wheel slip	None	Estimated	Computed
Wheel spin	No	No	Yes
Wheel lock	No	No	Yes
TCS/ABS testing	No	Limited	Full
Computational cost	Low	Low	Medium

Table 1: Tire model comparison

2 Force-Based Longitudinal Dynamics

2.1 Equations of Motion

Newton's second law in the longitudinal direction:

$$m \frac{dv}{dt} = \sum F_x - F_{\text{drag}} - F_{\text{roll}} \quad (4)$$

where:

$$\sum F_x = F_{x,\text{fl}} + F_{x,\text{fr}} + F_{x,\text{rl}} + F_{x,\text{rr}} \quad (\text{sum of tire forces}) \quad (5)$$

$$F_{\text{drag}} = \frac{1}{2} \rho C_d A v^2 \quad (6)$$

$$F_{\text{roll}} = C_{\text{roll}} \cdot \text{sgn}(v) \quad (7)$$

2.2 Tire Force Allocation

For a rear-wheel-drive vehicle:

$$F_{x,\text{fl}} = 0 \quad (\text{front wheels: unpowered}) \quad (8)$$

$$F_{x,\text{fr}} = 0 \quad (9)$$

$$F_{x,\text{rl}} = f_{\text{Dugoff}}(\sigma_{x,\text{rl}}, \sigma_{y,\text{rl}}, F_{z,\text{rl}}) \quad (10)$$

$$F_{x,\text{rr}} = f_{\text{Dugoff}}(\sigma_{x,\text{rr}}, \sigma_{y,\text{rr}}, F_{z,\text{rr}}) \quad (11)$$

where f_{Dugoff} is the tire model function (detailed in Section 4).

3 Normal Load Distribution

3.1 Static Load Distribution

Weight distribution between front and rear axles:

$$F_{z,\text{front}}^{\text{static}} = \frac{mg \cdot L_r}{L} \quad (12)$$

$$F_{z,\text{rear}}^{\text{static}} = \frac{mg \cdot L_f}{L} \quad (13)$$

where:

- m – vehicle mass (kg)
- $g = 9.81 \text{ m/s}^2$ – gravitational acceleration
- $L = L_f + L_r$ – wheelbase (m)
- L_f – distance from CoG to front axle (m)
- L_r – distance from CoG to rear axle (m)

3.2 Longitudinal Load Transfer

During acceleration or braking, normal load shifts between axles:

$$\boxed{\Delta F_z = \frac{m \cdot a_x \cdot h_{\text{cg}}}{L}} \quad (14)$$

where:

- a_x – longitudinal acceleration (m/s²), positive = acceleration
- h_{cg} – center of gravity height (m)

Dynamic normal loads:

$$F_{z,front} = F_{z,front}^{static} - \Delta F_z \quad (15)$$

$$F_{z,rear} = F_{z,rear}^{static} + \Delta F_z \quad (16)$$

Physical interpretation:

- $a_x > 0$ (acceleration): Weight transfers to rear $\Rightarrow$ more rear traction
- $a_x < 0$ (braking): Weight transfers to front $\Rightarrow$ more front braking capacity

3.3 Lateral Load Transfer

During cornering with lateral acceleration a_y :

$$\Delta F_{z,lat} = \frac{m \cdot a_y \cdot h_{cg}}{W} \quad (17)$$

where W is the track width (m).

3.4 Per-Wheel Normal Loads

Combined longitudinal and lateral load transfer:

$$F_{z,fl} = \frac{1}{2} (F_{z,front} - \Delta F_{z,lat,front}) \quad (18)$$

$$F_{z,fr} = \frac{1}{2} (F_{z,front} + \Delta F_{z,lat,front}) \quad (19)$$

$$F_{z,rl} = \frac{1}{2} (F_{z,rear} - \Delta F_{z,lat,rear}) \quad (20)$$

$$F_{z,rr} = \frac{1}{2} (F_{z,rear} + \Delta F_{z,lat,rear}) \quad (21)$$

3.5 Example: XCMG XDE320 Mining Truck

Parameters:

$$m = 218,000 \text{ kg (empty)} \quad (22)$$

$$L = 6.3 \text{ m} \quad (23)$$

$$h_{cg} = 2.8 \text{ m} \quad (24)$$

$$L_f = 2.52 \text{ m (40\% of wheelbase)} \quad (25)$$

$$L_r = 3.78 \text{ m (60\% of wheelbase)} \quad (26)$$

Static loads:

$$F_{z,front}^{static} = \frac{218,000 \times 9.81 \times 3.78}{6.3} = 1,284,000 \text{ N} \quad (27)$$

$$F_{z,rear}^{static} = \frac{218,000 \times 9.81 \times 2.52}{6.3} = 856,000 \text{ N} \quad (28)$$

Load transfer during braking at $a_x = -2.0 \text{ m/s}^2$:

$$\Delta F_z = \frac{218,000 \times (-2.0) \times 2.8}{6.3} = -193,600 \text{ N} \quad (29)$$

$$F_{z,front} = 1,284,000 + 193,600 = 1,478,000 \text{ N} \quad (30)$$

$$F_{z,rear} = 856,000 - 193,600 = 662,000 \text{ N} \quad (31)$$

Result: Front axle load increases by 15%, rear decreases by 23% during braking.

4 Dugoff Tire Model

4.1 Fundamental Equations

Tire forces are given by:

$$\boxed{F_x = C_x \sigma_x f(\lambda)} \quad (32)$$

$$\boxed{F_y = C_y \sigma_y f(\lambda)} \quad (33)$$

where:

- F_x, F_y – longitudinal and lateral tire forces (N)
- C_x, C_y – slip stiffness coefficients (N), load-dependent
- σ_x, σ_y – longitudinal and lateral slip ratios
- $f(\lambda)$ – friction saturation function
- λ – friction utilization parameter

4.2 Slip Ratios

Longitudinal slip ratio:

$$\boxed{\sigma_x = \frac{\omega R - V_x}{\max(|V_x|, |\omega R|, V_{\min})}} \quad (34)$$

where:

- ω – wheel angular velocity (rad/s)
- R – tire effective radius (m)
- V_x – vehicle longitudinal velocity (m/s)
- $V_{\min} \approx 0.1$ m/s – prevents division by zero

Sign convention:

$$\sigma_x > 0 \quad \text{Driving (wheel faster than vehicle)} \quad (35)$$

$$\sigma_x < 0 \quad \text{Braking (wheel slower than vehicle)} \quad (36)$$

$$\sigma_x = 0 \quad \text{Free rolling} \quad (37)$$

Lateral slip ratio:

$$\sigma_y = \frac{V_y}{V_x} \approx \tan(\alpha) \approx \alpha \quad (38)$$

where α is tire slip angle (rad), valid for small angles ($|\alpha| < 0.2$ rad).

4.3 Load-Dependent Slip Stiffness

Tire stiffness scales with normal load:

$$C_x(F_z) = C_{x,\text{base}} \cdot \frac{F_z}{F_{z,\text{ref}}} \quad (39)$$

$$C_y(F_z) = C_{y,\text{base}} \cdot \frac{F_z}{F_{z,\text{ref}}} \quad (40)$$

where:

- $C_{x,\text{base}}, C_{y,\text{base}}$ – baseline stiffness at reference load (N)
- $F_{z,\text{ref}}$ – reference normal load (N)

Typical values for mining truck tires:

- $C_{x,\text{base}} \approx 280,000$ N
- $C_{y,\text{base}} \approx 220,000$ N
- $F_{z,\text{ref}} = 100,000$ N

4.4 Friction Utilization Parameter

The key parameter governing saturation:

$$\lambda = \frac{\mu F_z}{2\sqrt{(C_x \sigma_x)^2 + (C_y \sigma_y)^2}} \quad (41)$$

where μ is the coefficient of friction (surface-dependent).

Physical interpretation:

$$\lambda \geq 1 \quad \text{Linear regime (tire not saturated)} \quad (42)$$

$$\lambda < 1 \quad \text{Nonlinear regime (friction-limited, sliding)} \quad (43)$$

4.5 Friction Saturation Function

$$f(\lambda) = \begin{cases} (2 - \lambda)\lambda, & \lambda < 1 \\ 1, & \lambda \geq 1 \end{cases} \quad (44)$$

Properties:

- Continuous at $\lambda = 1$: $\lim_{\lambda \rightarrow 1^-} f(\lambda) = 1$
- Smooth transition from linear to saturated regime
- Maximum value: $f_{\text{max}} = 1$ at $\lambda = 1$

4.6 Friction Circle Constraint

Theorem: The total tire force never exceeds the friction limit:

$$\sqrt{F_x^2 + F_y^2} \leq \mu F_z \quad (45)$$

Proof: For $\lambda < 1$ (friction-limited case):

$$\sqrt{F_x^2 + F_y^2} = \sqrt{(C_x \sigma_x f)^2 + (C_y \sigma_y f)^2} \quad (46)$$

$$= f(\lambda) \sqrt{C_x^2 \sigma_x^2 + C_y^2 \sigma_y^2} \quad (47)$$

$$= (2\lambda - \lambda^2) \cdot \frac{\mu F_z}{2\lambda} \quad (\text{from definition of } \lambda) \quad (48)$$

$$= \left(1 - \frac{\lambda}{2}\right) \mu F_z \quad (49)$$

$$< \mu F_z \quad (\text{since } 0 < \lambda < 1) \quad \square \quad (50)$$

4.7 Slip-Force Characteristic Curve

The Dugoff model produces the characteristic slip-force curve:

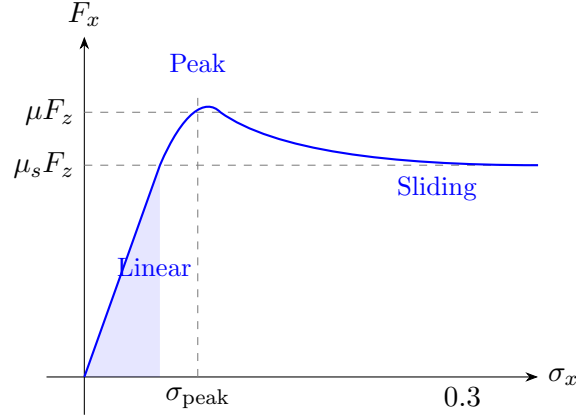


Figure 1: Dugoff slip-force characteristic showing linear region, peak grip, and sliding region

5 Wheel Rotational Dynamics

This section introduces the key addition for the hybrid model: independent wheel speed integration.

5.1 Motivation

In the torque-only model, wheel speed is derived from vehicle speed:

$$\omega = \frac{V_x}{R} \quad \Rightarrow \quad \sigma_x = 0 \text{ always!} \quad (51)$$

This prevents modeling of:

- Wheel spin during aggressive acceleration
- Wheel lock during hard braking
- Natural slip development
- TCS/ABS algorithm testing

The solution is to integrate wheel angular velocity independently.

5.2 Single Wheel Free Body Diagram

For each wheel, applying Newton's second law for rotation:

$$\boxed{I_w \dot{\omega} = \tau_{\text{drive}} - \tau_{\text{brake}} - F_x \cdot R} \quad (52)$$

where:

- I_w = wheel rotational inertia ($\text{kg} \cdot \text{m}^2$)
- ω = wheel angular velocity (rad/s)
- $\dot{\omega}$ = wheel angular acceleration (rad/s^2)

- τ_{drive} = drive torque at wheel (N·m)
- τ_{brake} = brake torque (N·m), always ≥ 0
- F_x = longitudinal tire force from Dugoff (N)
- R = tire effective rolling radius (m)

5.3 Drive Torque Distribution

For a rear-wheel drive (RWD) configuration:

$$\tau_{\text{drive,rl}} = \tau_{\text{drive,rr}} = \frac{\tau_{\text{motor}} \cdot N_{\text{gear}} \cdot \eta_{\text{drivetrain}}}{2} \quad (53)$$

$$\tau_{\text{drive,fl}} = \tau_{\text{drive,fr}} = 0 \quad (\text{front wheels not driven}) \quad (54)$$

where:

- τ_{motor} = motor shaft torque (N·m)
- N_{gear} = final drive gear ratio (dimensionless)
- $\eta_{\text{drivetrain}}$ = drivetrain efficiency (typically 0.90-0.95)

5.4 Wheel Inertia

The wheel rotational inertia includes:

$$I_w = I_{\text{tire}} + I_{\text{rim}} + I_{\text{brake}} + I_{\text{hub}} \quad (55)$$

Typical values:

- Passenger car: $I_w \approx 1.5 - 3.0 \text{ kg}\cdot\text{m}^2$
- Mining haul truck (XCMG XDE320): $I_w \approx 800 - 1500 \text{ kg}\cdot\text{m}^2$

5.5 Discrete-Time Integration

Using forward Euler integration:

$$\omega[k+1] = \omega[k] + \frac{\Delta t}{I_w} (\tau_{\text{drive}} - \tau_{\text{brake}} \cdot \text{sgn}(\omega[k]) - F_x[k] \cdot R) \quad (56)$$

Wheel Lock Detection:

$$\text{if } \omega[k] > 0 \text{ and } \omega[k+1] < 0 \text{ and } \tau_{\text{brake}} > 0 : \quad \omega[k+1] = 0 \quad (57)$$

This prevents non-physical sign reversal during braking.

5.6 Stability Criterion

The explicit integration is stable when:

$$\Delta t < \frac{2I_w}{C_x R^2} \quad (58)$$

For XCMG XDE320 parameters ($I_w = 1000 \text{ kg}\cdot\text{m}^2$, $C_x = 280,000 \text{ N}$, $R = 1.93 \text{ m}$):

$$\Delta t_{\text{max}} \approx \frac{2 \times 1000}{280,000 \times 1.93^2} \approx 1.9 \text{ ms} \quad (59)$$

Simulation timestep of 1 ms provides adequate stability margin.

6 Closed-Loop System Architecture

6.1 System Block Diagram

The hybrid model forms a closed-loop system:

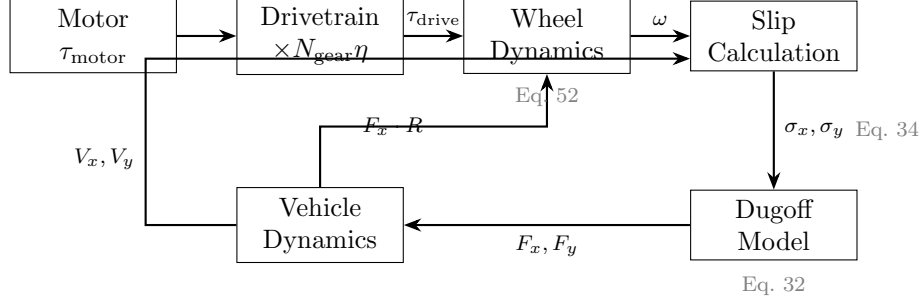


Figure 2: Closed-loop system architecture for hybrid torque-slip model

6.2 Causality Chain

At each timestep k :

1. **Input:** Motor torque command τ_{motor}
2. **Drivetrain:** $\tau_{\text{wheel}} = \tau_{\text{motor}} \cdot N_{\text{gear}} \cdot \eta$
3. **Wheel dynamics:** Integrate $\omega[k+1]$ using Eq. 56
4. **Slip calculation:** Compute σ_x, σ_y using Eq. 34
5. **Dugoff model:** Compute F_x, F_y using Eq. 32, 33
6. **Vehicle dynamics:** Update V_x, V_y from tire forces
7. **Feedback:** Tire reaction torque $F_x \cdot R$ affects next wheel integration

6.3 Key Insight: Feedback Loop

The tire force F_x appears on **both sides** of the system:

- **Output:** Accelerates the vehicle ($m\dot{V}_x = \sum F_x$)
- **Feedback:** Decelerates the wheel ($I_w\dot{\omega} = \tau - F_x R$)

This creates natural self-regulation:

- High slip $\rightarrow$ reduced F_x (saturation) $\rightarrow$ wheel accelerates less
- Low slip $\rightarrow$ high F_x $\rightarrow$ wheel slows toward free-rolling

7 Lateral Velocity Computation

7.1 Front Axle Lateral Velocity

Lateral velocity at front tire contact patch:

$$V_{y,\text{front}} = V_y + \dot{\psi} \cdot L_f - V_x \tan(\delta) \quad (60)$$

where:

- V_y – vehicle lateral velocity at CoG (m/s)
- $\dot{\psi}$ – yaw rate (rad/s)
- δ – steering angle (rad)

7.2 Rear Axle Lateral Velocity

$$V_{y,\text{rear}} = V_y - \dot{\psi} \cdot L_r \quad (61)$$

7.3 Lateral Slip Ratio per Wheel

$$\sigma_{y,\text{fl}} = \sigma_{y,\text{fr}} = \frac{V_{y,\text{front}}}{V_x} \quad (62)$$

$$\sigma_{y,\text{rl}} = \sigma_{y,\text{rr}} = \frac{V_{y,\text{rear}}}{V_x} \quad (63)$$

8 Torque-Only vs Hybrid Model

8.1 Torque-Only Model (Current Default)

Direct force computation without wheel dynamics:

$$F_x = \min \left(\frac{\tau_{\text{motor}} \cdot N_{\text{gear}} \cdot \eta}{R}, \mu F_z \right) \quad (64)$$

Advantages:

- Simple, fast computation
- No additional state variables
- Stable at all speeds
- Captures traction limiting

Limitations:

- Cannot model wheel spin ($\omega \gg V_x/R$)
- Cannot model wheel lock ($\omega = 0$ while $V_x > 0$)
- Slip ratio is estimated, not computed
- No TCS/ABS testing capability

8.2 Hybrid Model

Full causality: $\tau_{\text{motor}} \rightarrow \omega \rightarrow \sigma_x \rightarrow F_x$

Advantages:

- Physical wheel spin and lock behavior
- Natural slip development
- Full Dugoff slip-force curve (including peak and sliding)
- TCS/ABS algorithm testing

- Realistic launch and braking dynamics

Limitations:

- Additional state variables (4 wheel speeds)
- Requires smaller timestep for stability
- More complex initialization

8.3 Model Selection Guide

Use Case	Torque-Only	Hybrid
Path following	✓	✓
Energy estimation	✓	✓
Traction limiting	✓	✓
Wheel spin detection		✓
ABS/TCS development		✓
Stability control		✓
Loss of control		✓

Table 2: Model selection based on use case

9 Implementation Algorithm

9.1 Torque-Only Algorithm

Step 1: Compute Normal Loads

1. Static distribution from geometry
2. Load transfer: $\Delta F_z = ma_x h_{cg} / L$
3. Per-wheel loads

Step 2: Compute Demanded Force

$$F_{x,\text{demanded}} = \frac{\tau_{\text{motor}} \cdot N_{\text{gear}} \cdot \eta}{R} \quad (65)$$

Step 3: Apply Friction Limit

$$F_x = \min(|F_{x,\text{demanded}}|, \mu F_z) \cdot \text{sgn}(F_{x,\text{demanded}}) \quad (66)$$

Step 4: Update Vehicle Dynamics

$$a = \frac{\sum F_x - F_{\text{drag}} - F_{\text{roll}}}{m} \quad (67)$$

9.2 Hybrid Algorithm

Step 1: Compute Normal Loads (same as torque-only)

Step 2: Compute Slip Ratios from Current Wheel Speeds

$$\sigma_x = \frac{\omega R - V_x}{\max(|V_x|, |\omega R|, V_{\text{min}})} \quad (68)$$

Step 3: Compute Tire Forces (Dugoff)

$$\lambda = \frac{\mu F_z}{2\sqrt{(C_x \sigma_x)^2 + (C_y \sigma_y)^2}} \quad (69)$$

$$f(\lambda) = \begin{cases} (2 - \lambda)\lambda & \lambda < 1 \\ 1 & \lambda \geq 1 \end{cases} \quad (70)$$

$$F_x = C_x \sigma_x f(\lambda) \quad (71)$$

Step 4: Update Wheel Dynamics

$$\omega[k+1] = \omega[k] + \frac{\Delta t}{I_w} (\tau_{\text{drive}} - \tau_{\text{brake}} \cdot \text{sgn}(\omega) - F_x R) \quad (72)$$

Step 5: Update Vehicle Dynamics

$$V_x[k+1] = V_x[k] + \frac{\Delta t}{m} (\sum F_x - F_{\text{drag}} - F_{\text{roll}}) \quad (73)$$

9.3 Complete Discrete-Time State Update

$$\omega_i[k+1] = \omega_i[k] + \Delta t \cdot \dot{\omega}_i \quad (\text{for each wheel } i) \quad (74)$$

$$V_x[k+1] = V_x[k] + \Delta t \cdot a_x \quad (75)$$

$$\dot{\psi}[k] = \frac{V_x[k]}{L} \tan(\delta[k]) \quad (\text{or from lateral dynamics}) \quad (76)$$

$$\psi[k+1] = \psi[k] + \Delta t \cdot \dot{\psi}[k] \quad (77)$$

$$x[k+1] = x[k] + \Delta t \cdot V_x[k] \cos(\psi[k]) \quad (78)$$

$$y[k+1] = y[k] + \Delta t \cdot V_x[k] \sin(\psi[k]) \quad (79)$$

10 Surface Friction Coefficients

10.1 Mining Haul Road Conditions

Surface	μ_{peak}	μ_{slide}
Dry pavement	0.85	0.70
Compact gravel	0.72	0.60
Loose gravel	0.55	0.45
Iron ore dust (dry)	0.45	0.35
Iron ore dust (wet)	0.30	0.22
Mud	0.25	0.18
Sand (loose)	0.40	0.32

Table 3: Friction coefficients for Pilbara mining conditions

10.2 Effect of Surface Friction

The surface friction μ directly affects:

1. **Maximum traction force:** $F_{x,\text{max}} = \mu F_z$
2. **Maximum acceleration:** $a_{\text{max}} = \mu g$ (flat road)
3. **Stopping distance:** $d_{\text{stop}} = \frac{v_0^2}{2\mu g}$
4. **Friction utilization:** Lower μ means earlier saturation ($\lambda < 1$)

11 Implementation Parameters

11.1 XCMG XDE320 Mining Truck

Parameter	Value	Unit
Mass (empty)	218,000	kg
Mass (loaded)	538,000	kg
Wheelbase L	6.3	m
Track width W	4.0	m
CG height h_{cg}	2.8	m
Tire radius R	1.93	m
Gear ratio N_{gear}	9.0	-
Drivetrain efficiency η	0.92	-
Motor max torque	145,000	N·m
Motor max power	2,100	kW

Table 4: Vehicle parameters

Parameter	Value	Unit
$C_{x,base}$	280,000	N
$C_{y,base}$	220,000	N
$F_{z,ref}$	100,000	N
I_w (front)	800	kg·m ²
I_w (rear)	1,200	kg·m ²

Table 5: Tire and wheel parameters

12 Validation and Benchmarks

12.1 Expected Behavior

Scenario 1: Acceleration on gravel ($\mu = 0.72$)

XCMG XDE320 empty (218 tons):

$$F_{z,rear} \approx 0.60 \times 218,000 \times 9.81 = 1,283,000 \text{ N} \quad (80)$$

$$F_{x,max} = 0.72 \times 1,283,000 = 924,000 \text{ N} \quad (81)$$

$$a_{max} = \frac{924,000}{218,000} = 4.2 \text{ m/s}^2 \quad (82)$$

Scenario 2: Acceleration on wet dust ($\mu = 0.30$)

$$F_{x,max} = 0.30 \times 1,283,000 = 385,000 \text{ N} \quad (83)$$

$$a_{max} = \frac{385,000}{218,000} = 1.8 \text{ m/s}^2 \quad (84)$$

Result: Acceleration limited to 1.8 m/s² on wet dust vs 4.2 m/s² on gravel.

12.2 Stopping Distance Comparison

For $v_0 = 50 \text{ km/h} = 13.9 \text{ m/s}$:

$$d_{\text{gravel}} = \frac{13.9^2}{2 \times 0.72 \times 9.81} = 13.7 \text{ m} \quad (85)$$

$$d_{\text{wet dust}} = \frac{13.9^2}{2 \times 0.30 \times 9.81} = 32.8 \text{ m} \quad (86)$$

Stopping distance increases $2.4\times$ on wet dust compared to gravel.

13 Future Enhancements

13.1 Short-Term (V2.5)

- ✓ Wheel rotational dynamics (this document)
- ✓ Hybrid torque-slip model
 - Combined longitudinal-lateral friction circle

13.2 Medium-Term (V3)

- 3-DOF vehicle dynamics (add $V_y, \dot{\psi}$ as states)
- Oversteer/understeer simulation
- Loss of control scenarios

13.3 Long-Term (V4)

- 6-DOF dynamics (roll, pitch, heave)
- Suspension dynamics
- Rollover detection and prevention

14 Conclusion

This document presents a comprehensive vehicle dynamics model with two levels of tire modeling:

Torque-Only Model:

- Simple friction-limited force calculation
- Suitable for energy estimation and path following
- Cannot model wheel spin or lock

Hybrid Torque-Slip Model:

- Independent wheel speed integration
- Natural slip development
- Full Dugoff slip-force characteristic
- Enables TCS/ABS/ESC development

Both models share:

- Load transfer during acceleration/braking
- Surface friction effects
- Friction circle constraint

This forms a solid foundation for autonomous mining vehicle simulation.

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